

- Adv Disadv

Both are strong individually

In ans, we'll always take either adv or dis adv is strong

- If $\frac{s}{s}$ then choose 'Both'

IV Statement & Course of Action

→ The next step to → solve a Problem or minimize a Problem

- The soln shouldn't give others Problems.
- never choose an action whose outcome we can never predict

III Visual / Spatial reasoning

→ Translation
→ Scaling
→ Rotation
→ mirroring
→ Assembling
→ Grouping

→ Paper cutting
→ Paper folding
→ Pattern in 2D & 3D

- take 1 point, & the translation for that 1 point is applied to all, automatically & that's the same image.
points
- Apply for line → get the Scaling factors & automatically it'll be Scale factors for others lines too. (AS it's the same img).
- Scale factor = $\frac{\text{Post img}}{\text{Corresponding Pre img}}$

• General years

1984

1942

If last 2 digit %

4 == 0

Leap Years

• Century years

1600

2400

If Century Year % 400 == 0

Ly?

• B/W 2 century yrs there are 3 non-century

leap years.

eg - 1600 — Ly? ✓

1700 — x

1800 — x

1900 — x

2000 — ✓

} 3 non ly

For

• Does a leap year come after every 4 years?

No

eg → 1896 — ✓ 1900 — x 1904 — ✓

but why so,

Then now we know → 365 + (6 hrs) → Ly

This

makes

diff. over

100 years

But it's not 6 hrs, it's 5 hrs 48 min sec

1600, 1800

2000

Century 1/14

so

no. of years	Ly	Ny
100 →	24	76
200 →	48	152
300 →	72	228
400 →	97	303
	$2 \times 4296 + 1$	$76 \times 4 = 304 - 1$

$(1 - 100) / 4 = 25 \text{ Ny} - 1 = 24 \text{ Ly}$
 $\swarrow$
 $75 \text{ Ny} + 1 = 76 \text{ Ny}$
 Century

$\therefore \frac{100}{400}$, rem $\neq 0 \therefore$ not Ly but Ny

• odd days

no. of days	Day of the week
0	Sun
1	Mon
2	Tue
3	Wed
4	Thurs
5	Fri
6	Sat

eg 10 days $\rightarrow 1 \text{ w} + 3 \text{ d}$
 Jan (31) $\rightarrow 4 \text{ w} + 3 \text{ d}$
 365 d $\rightarrow 52 \text{ w} + 1 \text{ day}$

No. of odd days in a century

$$365 \rightarrow 52W + 1$$

$$366 \rightarrow 52W + 2$$

No. of years	LY	NY	No. of odd days
100 $\rightarrow$	24	76	$\rightarrow \frac{24 \times 2 + 76}{7} = \frac{124}{7} = 17W + 2$
200 $\rightarrow$	48	152	$\rightarrow 3W + 3$
300 $\rightarrow$	72	228	$\rightarrow 3W + 1$
400 $\rightarrow$	94	303	$\rightarrow 7W + 0$
500 $\rightarrow$			5
600 $\rightarrow$			3
700 $\rightarrow$			1
800 $\rightarrow$			0

} Same Pattern

• When it's a century leap year (400, 800 $\rightarrow$)

$\rightarrow$ 0 no. of odd days

• Year No. of odd days

1600 $\rightarrow$	0
2000 $\rightarrow$	0
2400 $\rightarrow$	0

• A calendar repeats in 400 yrs.

• When the no. of odd days become 0, the calendar repeats.

• If the year is LY, then the same/alike nearest year will also be LY, if NY then NY; otherwise for odd days, it will make diff.

• If x^{th} Jan Z^{th} (year) was 'y' day. Then what was the day of the week on x^{th} Jan Z^{th} (year)?

No. of odd days coming pending to 'y' day. - If Z^{th} is NY, odd; if LY, odd

12 Clocks

- The current time in mirror / actual clock =

11:60 - whatever the time it is in actual / mirror clock



1 hr $\rightarrow$ 5 ^{min} Spaces

12 $\rightarrow$ 60

min Spaces

- From 12am-12pm $\Rightarrow$ There are 60 min Spaces in a clock
- min hand moves 1 min $\rightarrow$ That means, min hand moves 1 min Spaces

like this

60 min $\rightarrow$ 60 min / 1 hr

- In 1 hr, min hand goes 360 in 1 hr

In 1 hr, hr hand goes 30°

In 1 min, sec hand goes 360°

In 1 min, min hand goes 6°

In 1 min, hr hand moves 20.5°

In 1 hr, the min hand goes/gains 55 min Spaces over

the hr hand.

- Whenever it's 3, 6, 9 etc $\rightarrow$ the angle is 90°
- Whenever it's 30 $\rightarrow$ there are 15 min Spaces

- Whenever we want an angle of 90°, the min & hr hands will be 15 min Spaces apart.

- Every 1 hr we can find 90° angle for 2 times

- --- $[12 \text{ hrs} \text{ --- } 360^\circ \text{ angle}] = [22 \text{ times}]$

- --- $[24 \text{ hrs} \text{ --- } 720^\circ] = [44 \text{ times}]$

- For every 1 hr, the min & hr hand coincide for 1 time.

- For 12 hrs. --- = 11 times

- For 24 hrs. --- = 22 times

- [Coinciding $\rightarrow 0^\circ$], [opposite to 180°]

- In 1 hr, min & hr hands together make 180° for 1 time

- In 12 hrs. --- $[180^\circ = 11 \text{ times}]$

- In 24 hrs. --- $[180^\circ] = 22 \text{ times}$

- In 1 day there will be 22 times coinciding & 22 times facing opposition of min & hand.

- Both for coincide & facing opposite, the min & hr hands are in straightline.

- In 1 day for 44 times $(22 + 22)$ both hands will be in straightline (22 for C., 22 for opposite.)

- C. t. = 12:10 pm, other clock = 12:15. this means this clock has gained 5 more mins / ahead of by 5 mins.

- C. t. = 12:10 pm, other clock = 12:05, --- [behind by 5 mins]

- Fast clock (gaining mins) $\rightarrow$ Gains x mins of time / hr

- Slow clock (losing mins) $\rightarrow$ Loses x mins of time / hr

- A broken clock gives correct time, twice / day.

- $\swarrow$ exception (might be)
Fast / Slow clock $\rightarrow$ it can be 2/1 not at all gives correct time (might)

• If a slow-fast clock loses/gains 12 hrs then,
It shows the correct time at least once.

• Eg 9:15 → At what time b/w 12 am to 1 pm the
clock hands will be together / in opposite direction?

↓
For these type of Qns, the movement of hr hand
will not be considered i.e. hr hand is fixed.

• After $65 \frac{5}{11}$ mins the hr & min hands will always
be together

• If a clock gains time → the alarm ^{will} ring before
expected time
• loses time → after
expected time